# EXPERIMENT-14: EXPLORING PRO-CODE EXTENSION OF A ZOHO CREATOR APPLICATION USING ZOHO CATALYST / CUSTOM FUNCTIONS

## AIM

To extend the Event Registration Management System in Zoho Creator using custom Deluge functions and, where required, Zoho Catalyst serverless services for advanced computation, external API integration, and intensive processing.

## THEORY

Zoho Creator provides forms, reports, workflows, and Deluge scripting for rapid application development. More complex event registration requirements may benefit from reusable custom functions or pro-code backend services.

### 1. Custom Deluge Functions

Custom Deluge functions encapsulate reusable business logic inside Zoho Creator. In the Event Registration Management System, they may be used to validate participant details, check event capacity, update registration status, and call external REST APIs.

### 2. Zoho Catalyst Serverless Pro-Code Integration

Zoho Catalyst supports serverless backend development using languages such as Java, Node.js, and Python. If the event registration application requires processing unsuitable for a simple Creator workflow, such as handling QR ticket generation, sending event notifications through an external service, or processing large registration batches, the work can be handled by a Catalyst function and called through an authenticated API.

## PROCEDURE

### Step 1: Create a Custom Deluge Function

1. Open the Event Registration Management System in Zoho Creator.
2. Navigate to the application logic/functions area available in your Creator interface.
3. Choose Create Function, then define a suitable name, input arguments, and return type.
4. Write and test the Deluge logic, then save the function.
5. Associate the function with the relevant event registration workflow, such as after a record is submitted or updated.

### Example Deluge Logic — Registration / Admission Validation

```
if(input.Event_Capacity > 0)
{
    if(input.Registered_Count < input.Event_Capacity)
    {
        input.Registration_Status = "Confirmed";
    }
    else
    {
        input.Registration_Status = "Waitlisted";
    }
}
```

Use the actual field link names from your application. If the function is called from a workflow, pass or reference the relevant field values according to the function configuration.

### Step 2: Set Up a Zoho Catalyst Project (Optional)

1. Open the Zoho Catalyst Console and create a project if a separate pro-code backend is required.
2. Use the supported development environment or CLI to create a serverless function in Node.js, Java, or Python.
3. Implement the advanced task required by the event registration, such as handling QR ticket generation, sending event notifications through an external service, or processing large registration batches.
4. Deploy the function and note its API endpoint and authentication requirements.

### Step 3: Connect Creator to Catalyst Using an API

1. In Creator, open the Connections area under the integration settings and create an authenticated connection if your account supports it.
2. Configure the Catalyst endpoint and authentication securely. Do not expose API keys or tokens in a public form or hardcode sensitive credentials in shared scripts.
3. Use an invokeurl request in a custom function to send only the required event registration data to the endpoint.

4. Parse the response and use the returned result in the appropriate application field or report.

### Example Request Structure

```
response = invokeurl
[
    url : "YOUR_AUTHENTICATED_CATALYST_ENDPOINT"
    type : POST
    parameters : {"eventId": input.ID, "participantEmail": input.Email}
    connection : "your_connection_link_name"
];
```

### Step 4: Test and Validate the Workflow

1. Submit a test event registration record or use a safe test record.
2. Confirm that the Creator workflow triggers the custom function.
3. If Catalyst is used, verify that the request reaches the endpoint and returns the expected response.
4. Check the calculated or validated result in the event registration record and report.
5. Test invalid inputs and API failures so the application handles errors safely.

### RESULT

The pro-code extension capabilities of Zoho Creator were studied using the Event Registration Management System. Reusable Deluge functions can support registration validation and event-capacity checks. When advanced processing is required, Zoho Catalyst can provide a separate pro-code backend through authenticated API calls. Actual implementation depends on the field link names, connections, and features available in the Zoho Creator account.